

Electrical spectroscopy of porous rocks: a review—I. Theoretical models

T. L. Chelidze¹ and Y. Gueguen²

¹ *Institute of Geophysics, Georgian Ac. Sci. 1, Alexidze str. 380093, Tbilisi, Georgia and Geosciences Rennes, Université de Rennes 1, 35042, Rennes, France*

² *Ecole Normale Supérieure, Laboratoire de Géologie, Dep. TAO, 24, rue Lhomond, Paris, France. E-mail: gueguen@geologie.ens.fr*

Accepted 1998 August 25. Received 1998 July 28; in original form 1997 July 15

SUMMARY

The complex dielectric permittivity ϵ^* of porous water-bearing rocks in the frequency range from a few to hundreds of megahertz reveals several intensive relaxation effects and a non-trivial dependence on the water content. At high frequencies, $f > 10$ MHz, both the real part of the complex dielectric permittivity ϵ' and the conductivity σ of water-bearing rocks are correctly predicted by the Maxwell–Wagner–Bruggeman–Hanai (MWBH) theory of composite dielectrics. This theory takes into account only the bulk properties of components, their partial volumes and the configuration of particles. The theory ignores two important factors: the surface contribution to polarization and the effect of clustering of components. At frequencies $f < 10$ MHz there are certain frequency domains which exhibit relaxation processes not predicted by MWBH theory. The characteristic times of these processes range from 10^{-6} to 10 s. These relaxation effects are related to different surface polarization processes which are, in order of increasing water content, (i) orientational polarization of bound water, (ii) polarization of liquid films or pockets, producing a polarization catastrophe effect, (iii) polarization of rough fractal surfaces, (iv) polarization of the ‘closed’ electrical double layer (EDL), when the displacement of the excess surface charges is limited by the external boundary of the EDL, and (v) polarization of the ‘open’ double layer, implying free exchange of excess ions with the bulk electrolyte and generation of transient diffusional potentials, which lag behind the applied field. Some theoretical models predict large effective values of relative dielectric constants in the range 10^5 – 10^6 at low frequencies. Knowledge of the characteristic signatures of these physical mechanisms is important for the correct interpretation of experimental data.

Analysis of existing theories of polarization of heterogeneous media shows that electrical spectroscopy can be useful for the interpretation of frequency spectra of complex dielectric permittivity or conductivity of water-bearing rocks and porous materials in general, and for the determination of water content, its thermodynamic state, the connectivity of water-bearing channels and their correlation lengths and the surface to volume ratio and surface charge in particular, in addition to the traditional formation factor, which is obtained from ohmic conductivity measurements.

Key words: dielectric permittivity, electrical conductivity, electromagnetism, fractals, inhomogeneous media, polarization.

1 INTRODUCTION

Electrical properties of rocks are of great interest for both exploration of the deep Earth and exploration of the shallow crust. Due to the intensive development of magnetotelluric and alternating current methods, the electrical imaging of geological structures has become decisive in solving some

prospecting problems. The monitoring of electrical properties can be very useful for the control of pollution (Roberts & Lin 1997), oil reservoir monitoring, hydrothermal energy and earthquake prediction studies (Mogi 1985) and so on. Correct interpretation of field data calls for a clear understanding of possible sources of variations of electrical properties. This is a very delicate problem as electrical properties of porous media

such as conductivity, dielectric permittivity (DP) and electrochemical potentials are extremely sensitive to physical conditions (temperature, pressure), the chemical composition of solid and liquid phases, and microstructural factors (porosity, its distribution and its connectivity).

The main purpose of this paper is to discuss the principal physical models and to consider critically possible physical mechanisms of dielectric losses and polarization under various conditions in order to review and explain the principal experimental results obtained by methods of low-frequency alternating current spectroscopy of rocks (Chelidze *et al.* 1999, hereafter referred to as paper II). We restrict ourselves to the range of relatively weak applied fields, thus excluding non-linear effects, which are covered by the studies of Olhoeft (1981, 1985). The frequency domain of electrical spectroscopy considered in this paper covers the interval from 1 mHz to 100 MHz, thus we do not consider here the domain of ultra-high frequencies, which is considered by Hoekstra & Doyle (1971) and Shen *et al.* (1985).

Typical rock in a vacuum is a good insulator with a conductivity of the order of $10^{-10} \text{ S m}^{-1}$ or less, but brine-saturated samples can have conductivities of the order of 1 S m^{-1} . This makes the measurement of rock electric properties a difficult technical problem. In addition, the possible superposition of various types of conduction and polarization, both in volume and on the solid–liquid interface, complicates dielectric spectra and calls for model experiments, where the influence of a single factor can be traced reliably.

The principles of solid-state physics, the physics of disordered media and the physics and chemistry of surfaces are now intensively applied to the interpretation of electrical spectra of rocks. The first study in this area was that of Smith-Rose (1934), who was surprised to discover a strong dependence of the real part of complex dielectric permittivity ε^* on frequency, and very high values of DP at low frequencies for samples of wet rocks and soil, ε of the samples being larger than that of the components. He ascribed this effect to the granular structure of samples. This effect of large low-frequency polarization has recently been a subject of discussion: in general the question has been posed whether it is possible for a mixture of components to have greater DP than that of any of its components. Some authors explain high dielectric constant values of the order of thousands and more by electrode polarization. Electrode impedances can greatly distort the measured electrical parameters of rocks at low frequencies, and special methods of separation and exclusion of the electrode contribution to measured electrical characteristics have been developed (see Paper II). As a result, very high values of DP for water bearing rocks have been confirmed (Chelidze 1963). It is worth noting that the electrical behaviour of dry rocks and bulk water is quite trivial and follows well-known theories of solid dielectrics and polar liquids. At the same time, water-bearing rocks demonstrate unusual dielectric spectra, enormous static polarization and anomalously large relaxation times. All of these non-trivial effects result, as shown below from physical and chemical processes on the solid–liquid interface, and can be considered as surface contributions to the complex electrical conductivity of a porous system.

Contributions to ac electrical studies of rocks are presented in the papers of Keller & Licastro (1959), Howell & Licastro (1961), Parkhomenko (1967), Chelidze (1971), Chelidze *et al.* (1977, 1979), Olhoeft (1981, 1985), Keller (1982), Knight (1983,

1991), Lockner & Byerlee (1985), Sen (1989), Ruffet *et al.* (1991a,b), Knight & Dvorkin (1992), Ruffet (1993), Garrouch & Sharma (1994), Glover *et al.* (1994a,b, 1996), Roberts & Tyburczy (1994), etc.

2 GENERAL DEFINITIONS

Application of an electric field to a rock causes both irreversible migration and reversible displacement of electric charges. The first process is characterized by the specific conductivity σ , and the latter by the dielectric permittivity ε .

The polarization vector $\mathbf{P}$ of a material is defined as the induced electric dipole moment per unit volume. The electrical displacement vector $\mathbf{D}_e$ is related to the electric field $\mathbf{E}$ and to $\mathbf{P}$ by

$$\mathbf{D}_e = \varepsilon \mathbf{E} + \mathbf{P}, \quad (1)$$

where ε is the dielectric permittivity of the material. The quantities $\mathbf{E}$, $\mathbf{D}_e$, σ and ε (or, alternatively, ε') are expressed in SI units as follows: V m^{-1} , C m^{-2} , S m^{-1} and F m^{-1} , respectively (see Appendix A for symbols and abbreviations).

The total current $\mathbf{J}_t$ resulting from the applied voltage $V = V_0 \exp(i\omega t)$ is the sum of a conduction current $\mathbf{j}_c$ and a displacement current $\mathbf{j}_d$:

$$\mathbf{J}_t = \mathbf{J}_c + \mathbf{J}_d, \quad (2)$$

$$\mathbf{J}_c = \sigma \mathbf{E}, \quad (3)$$

$$\mathbf{J}_d = \varepsilon_d^* \partial \mathbf{E} / \partial t = i\omega \varepsilon_d^* \mathbf{E}, \quad (4)$$

$$\mathbf{J}_t = (\sigma + i\omega \varepsilon_d^*) \mathbf{E} = \sigma^* \mathbf{E}, \quad (5)$$

where ω is the angular frequency and $\varepsilon_d^* = (\varepsilon' - i\varepsilon'')$ is the complex dielectric permittivity *excluding* ohmic losses. The term ε_d'' is caused by energy dissipation during polarization, for example by the rotation of dipole molecules in a viscous solvent. Eq. (5) can be rewritten as

$$\mathbf{J}_t = i\omega \varepsilon^* \mathbf{E} \quad \text{or} \quad \mathbf{J}_t = \sigma^* \mathbf{E}, \quad (6)$$

where

$$\varepsilon^* = \varepsilon_d^* + \sigma / i\omega = \varepsilon' - i(\varepsilon_d'' + \sigma / \omega) = \varepsilon' - i\varepsilon'', \quad (7)$$

$$\sigma^* = \sigma' + i\sigma'' = i\omega \varepsilon^* = i\omega(\varepsilon' - i\varepsilon''). \quad (8)$$

The ratio of the imaginary part of ε^* to the real part $\varepsilon''/\varepsilon'$ (or the ratio σ''/σ') is defined as the loss angle tangent, $\tan \delta$:

$$\tan \delta = \varepsilon''/\varepsilon' = \sigma''/\sigma'. \quad (9)$$

The loss angle tangent due only to relaxation processes (see Fig. 1a) is

$$\tan \delta_d = \varepsilon_d''/\varepsilon'. \quad (10)$$

In the following text ε will stand for ε' in order to simplify expressions.

The electrical spectrum of a given material is defined as the plot of ε' and ε'' (or $\tan \delta$) versus frequency $f = \omega/2\pi$. In physical terms, it is the Fourier transformation of a transient process, namely of the induced polarization decay function. Polarization processes are quite often characterized by an exponential decay with a relaxation time τ , and maximal energy dissipation is observed at the critical frequency $f_c = 1/2\pi\tau$ (Debye 1929; Frölich 1990). In this case, which is called Debye relaxation, the complex dielectric permittivity is

$$\varepsilon^* = \varepsilon - i\varepsilon'' = \varepsilon_\infty + \frac{(\varepsilon_s - \varepsilon_\infty)}{(1 + i\omega\tau)} - i\frac{\sigma}{\omega}, \quad (11)$$

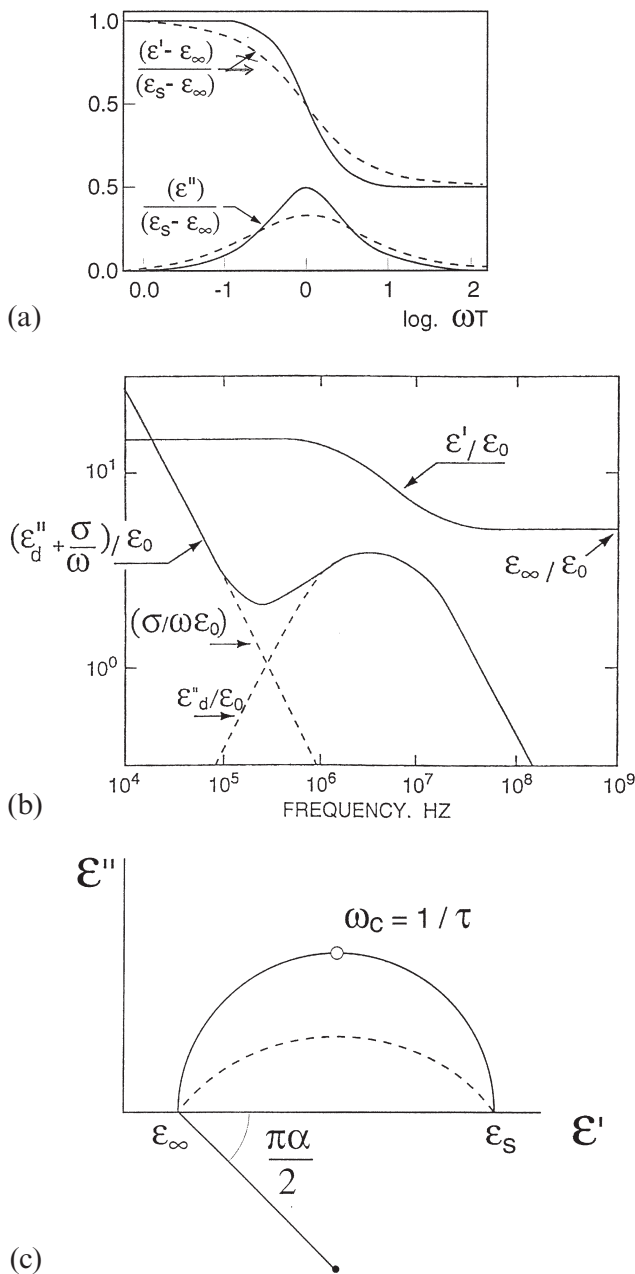


Figure 1. (a) The real and imaginary parts of relative dielectric permittivity (dielectric constant) for a rock with $\sigma = 0$. Solid curves: Debye model; dashed curves: Cole–Cole model (Gueguen & Palciauskas, 1994). (b) General behaviour of the real (ϵ'/ϵ_0) and imaginary ($\epsilon''_d/\epsilon_0 + \sigma/\omega\epsilon_0$) parts of the complex dielectric constant ϵ^*/ϵ_0 for a rock with ohmic conductivity σ . Higher saturation and conductivity of brine increase ϵ^*/ϵ_0 by a term $\sigma/\omega\epsilon_0$ (Gueguen & Palciauskas, 1994). (c) Cole–Cole plot of ϵ'' as a function of ϵ' for a rock with $\sigma = 0$. Solid line: a single relaxation time; dashed line: Cole–Cole distribution of relaxation times with $\alpha = 0.5$.

where ϵ_∞ and ϵ_s correspond to $f \gg f_c$ and $f \ll f_c$ respectively (Figs 1a and b).

In real dielectrics, polarization is the sum effect of many relaxation processes with different relaxation times, and instead of the Debye equation, an empirical expression, proposed by Cole & Cole (1941), is usually used (Fig. 1a):

$$\epsilon^* = \epsilon_{cc} - i\sigma/\omega, \quad (12)$$

with

$$\epsilon_{cc} = \epsilon_\infty + \frac{(\epsilon_s - \epsilon_\infty)}{1 + (i\omega\tau)^{1-\alpha}}, \quad (13)$$

where α is the Cole–Cole distribution function exponent. The plot of ϵ'' versus ϵ' can be more complicated (skewed arc), and this can also be taken into account by additional exponents in the Cole–Cole equation (Chelidze *et al.* 1977).

For a very broad distribution of relaxation times there is no dominant characteristic time constant, with all frequencies contributing to the polarization process. The scale-invariant relaxation manifests itself by an inverse power-law frequency spectrum of ϵ' and frequency-independent $\tan \delta$ (Hanai 1968; Clerc *et al.* 1990).

The type of relaxation time distribution functions can be easily determined from plots of ϵ'' versus ϵ' or σ' versus σ'' for a broad range of frequencies, which are known as Cole–Cole or Argand plots (Fig. 1c). The Cole–Cole plot for a single relaxation time is a semi-circle between $\epsilon' = \epsilon_\infty$ and $\epsilon' = \epsilon_s$, centred on the ϵ' -axis. For the symmetrical distribution of relaxation times around a dominant time constant, it is the section of a circle with its centre below the ϵ' -axis; the value $\pi\alpha/2$ is the angle between the ϵ' -axis and a straight line from the point $\epsilon' = \epsilon_\infty$ to the centre of the circle. Larger values of α represent broader relaxation time distribution functions. α can thus in principle characterize the degree of heterogeneity of the material.

Any type of frequency behaviour of ϵ' and ϵ'' can be readily mimicked by a properly arranged combination of resistances R and capacitances C , which is called an equivalent circuit. In particular, water-bearing rock can be considered as a network of bonds with complex resistances R^* associated with porous space, saturated partially or fully by the fluid; mineral grains are good dielectrics and thus the main conductive paths follow pores and cracks. A well-designed equivalent circuit can give some indication of the general model of conduction and polarization (Ruffet *et al.* 1991; Ruffet 1993; Roberts & Tyburczy 1994). Nevertheless, it should be borne in mind that different equivalent circuits can demonstrate practically identical frequency behaviour (Von Hippel 1954). This means that the equivalent circuit approach does not give a unique solution, and only physical experiments under various conditions can reveal the true physical mechanisms of conduction and polarization.

It is commonly assumed that ohmic conduction is an irreversible process and that dielectric polarization, on the other hand, can be caused only by the reversible displacement of charges. The situation is somewhat different in thermodynamically open systems, where various flows are generated by the application of a physical field (thermodynamic force). According to the thermodynamics of irreversible processes (De Groot & Mazur 1967), the interaction of flows produces some reversible contribution to the process such as electrical polarization. For example, interference of irreversible processes of migration and diffusion can produce electrical polarization in rocks, which in principle can explain such unusual effects as the simultaneous increase of both the dielectric constant and conductivity. Such phenomena are impossible to explain from the static approach as, according to this approach, the higher the ohmic conductivity of a system is, the less the chances for a local accumulation of charges are, i.e. for polarization. The detailed physical mechanism and kinetics of such processes are considered in Section 3.2.2.

3 THEORIES OF ELECTRIC PROPERTIES OF HETEROGENEOUS SYSTEMS

3.1 Bulk effects (Maxwell–Wagner, Bruggeman, Hanai)

3.1.1 First-order models: layers and dilute suspensions

Theories of effective electrical and other transport properties of composite media are founded on Maxwell's approach (Maxwell 1891). In fact, Maxwell derived the expression for the dielectric constant and conductivity of a two-layer dielectric, approximating it by imperfect capacitors (RC circuits in series). It was later extended to a dilute suspension of spheres by Wagner (1924) and to a suspension of ellipsoidal particles by Fricke (1924) and Sillars (1937).

The effective medium approach (EMA) can be formulated in the following way. Consider a mixture of two ideal dielectrics with dielectric permittivities ε_1 and ε_2 and partial volumes $\Phi_1 = V_1/V$ and $\Phi_2 = V_2/V$, where V is the total volume of the mixture and V_1 and V_2 are those of the first and the second components. The mean electric field in the mixture $\langle \mathbf{E} \rangle$ can be calculated as

$$\langle \mathbf{E} \rangle = (1/V) \int_v \mathbf{E} dv, \quad (14)$$

and those in the components as

$$\langle \mathbf{E}_1 \rangle = (1/V_1) \int_{v1} \mathbf{E}_1 dv, \quad (15)$$

$$\langle \mathbf{E}_2 \rangle = (1/V_2) \int_{v2} \mathbf{E}_2 dv. \quad (16)$$

Thus,

$$\langle \mathbf{E} \rangle = \Phi_1 \langle \mathbf{E}_1 \rangle + \Phi_2 \langle \mathbf{E}_2 \rangle. \quad (17)$$

Similarly, for the mean electrical displacement vector $\langle \mathbf{D}_e \rangle$,

$$\langle \mathbf{D}_e \rangle = \Phi_1 \langle \mathbf{D}_1 \rangle + \Phi_2 \langle \mathbf{D}_2 \rangle. \quad (18)$$

The effective dielectric permittivity of mixture ε is

$$\varepsilon = \langle \mathbf{D}_e \rangle / \langle \mathbf{E} \rangle, \quad (19)$$

and those of the components are

$$\varepsilon_1 = \langle \mathbf{D}_1 \rangle / \langle \mathbf{E}_1 \rangle, \quad (20)$$

$$\varepsilon_2 = \langle \mathbf{D}_2 \rangle / \langle \mathbf{E}_2 \rangle. \quad (21)$$

Introducing the field ratios f of the components as

$$f_1 = \langle \mathbf{E}_1 \rangle / \langle \mathbf{E} \rangle, \quad (22)$$

$$f_2 = \langle \mathbf{E}_2 \rangle / \langle \mathbf{E} \rangle, \quad (23)$$

it follows from the equations for displacement vector (18), dielectric permittivity (19–21) and field ratio (22, 23) that

$$\varepsilon = \varepsilon_1 + (\varepsilon_2 - \varepsilon_1) \Phi_2 f_2. \quad (24)$$

In the following sections we will assume that the mixture is a matrix with dielectric permittivity ε_1 , containing inclusions with dielectric permittivity ε_2 . We also assume that the size of the inclusions is much less than that of the sample so that the sample can be considered as macroscopically homogeneous. This is not true, however, in systems with a fractal structure, where the homogeneous regime is attained only when the sample size L is much greater than the correlation length L_c ;

that is, greater than the largest cluster of heterogeneities (Chelidze *et al.* 1998).

The main difficulty is in the calculation of the field ratio f_2 . Considering a dilute suspension of spheres, Wagner (1924) derived the field ratio as

$$f_2 = \frac{3\varepsilon_1}{2\varepsilon_1 + \varepsilon_2}. \quad (25)$$

Substituting eq. (25) into eq. (24), we obtain the simplest formula for a suspension of spheres:

$$\varepsilon = \varepsilon_1 + 3\varepsilon_1 \frac{\varepsilon_2 - \varepsilon_1}{2\varepsilon_1 + \varepsilon_2} \Phi_2. \quad (26)$$

The increment of dielectric permittivity of a matrix due to the addition of inclusions is,

$$\Delta\varepsilon = \varepsilon - \varepsilon_1 = 3\varepsilon_1 \frac{\varepsilon_2 - \varepsilon_1}{2\varepsilon_1 + \varepsilon_2} \Phi_2. \quad (27)$$

Similarly, f_2 can be calculated exactly for ellipsoidal inclusions (Reinolds & Hough 1957):

$$f_2 = \sum_{i=1}^3 \frac{\cos^2 \Theta_i}{1 + A_i[(\varepsilon_2/\varepsilon_1) - 1]}, \quad (28)$$

where A_i are three depolarization factors (Dukhin & Shilov 1974), Θ_i are three angles between the ellipsoidal axis and the field $\mathbf{E}$, and ε_2 and ε_1 are the dielectric permittivities of the inclusions and the matrix, respectively. For spheres, $A_i = 1/3$, $\cos^2 \Theta_i = 1/3$ and from eq. (28) we recover eq. (25).

The main difference between the numerous effective medium theories (Reinolds & Hough 1957; Hanai 1968; Chelidze *et al.* 1977) is the assumption made about the value of the dielectric constant of the embedding substance, ε_e , which differs from the dielectric constant of the matrix due to the interaction of the electrical fields of inclusions. The simplest assumption is $\varepsilon_e = \varepsilon_1$, as above (eq. 28); that is, the dielectric permittivity of the equivalent embedding medium is that of the first component (matrix). This approximation is valid only for very dilute mixtures and relatively small contrasts between ε_1 and ε_2 when the electrical fields of components do not overlap significantly (Gueguen *et al.* 1997). Other approximations are examined below.

3.1.2 Self-consistent and differential self-consistent models

Some authors (see Reinolds & Hough 1957; Chelidze *et al.* 1977) substitute in eq. (28) instead of $\varepsilon_e = \varepsilon_1$ the value of the effective dielectric permittivity of the mixture itself, that is $\varepsilon_e = \varepsilon$; this approach results in a so-called self-consistent model. However, it seems that the best way to take into account the interaction of the inclusions is a differential self-consistent approach, which has been developed by Bruggeman (1935). In this approach the final concentration of inclusions is approached step by step by addition of infinitesimal portions of inclusions $\delta\Phi$. The increment $\delta\varepsilon$ of dielectric permittivity of the mixture ε due to the n th addition of such a small portion of inclusions is then given correctly by (27):

$$\delta\varepsilon_n = \varepsilon - \varepsilon_n = 3\varepsilon_n \frac{\varepsilon_2 - \varepsilon_n}{2\varepsilon_n + \varepsilon_2} \delta\Phi_n, \quad (29)$$

which can be considered as a differential equation linking $\delta\varepsilon_n$ and $\delta\Phi_n$. After integration from $\varepsilon = \varepsilon_1$ and $\Phi_2 = 0$ to $\varepsilon_n = \varepsilon$ and

$\Phi_n = \Phi_2$, we obtain Bruggeman's formula:

$$\frac{\varepsilon - \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \left(\frac{\varepsilon_1}{\varepsilon} \right)^{1/3} = 1 - \Phi_2. \quad (30)$$

Hanai (1968) developed a comprehensive theory of the complex dielectric constant of a concentrated suspension of spheres, where each component is characterized by a complex dielectric constant ε_i^* , and derived the complex analogue of Bruggeman's equation,

$$\frac{\varepsilon^* - \varepsilon_2^*}{\varepsilon_1^* - \varepsilon_2^*} \left(\frac{\varepsilon_1^*}{\varepsilon^*} \right)^{1/3} = 1 - \Phi_2. \quad (30a)$$

This expression can be considered as the final result of the development of the effective media approach to the analysis of suspension-type composite dielectrics. We will refer to it as Maxwell–Wagner–Bruggeman–Hanai (MWBH) theory. From the complex equation (30a), the simple expressions for some practically important limiting cases, namely for low or high frequency, high conductivity of inclusions or high conductivity of matrix, can be obtained. For example, after cumbersome transformations of eq. (30a) in the low-frequency (LF) limit at $\sigma_2 \gg \sigma_1$, the real part of the complex dielectric permittivity of the mixture can be obtained as (Hanai 1968; Dukhin & Shilov 1974)

$$\varepsilon = \frac{\varepsilon_1}{(1 - \Phi_2)^3}, \quad (31)$$

from which we can conclude that conducting particles embedded in the dielectric cause a large increment of the dielectric permittivity as $\Phi_2 \rightarrow 1$. Thus, although water inclusions in a dielectric (say oil) at high frequencies (HF) have a standard value of $\varepsilon = 80$, at low enough frequencies the inclusions succeed in screening the external electric field due to the displacement of charges over a distance of the order of the Debye length, which means that in the LF limit the water inclusions in the dielectric have infinite dielectric permittivity (compare Figs 2a and b). Note that in the above formulae the ε^* and consequently σ^* of a suspension of spherical particles do not depend on the size of inclusions. Theories for ellipsoidal particles predict a strong dependence of ε^* on the aspect ratio, but again ε^* does not depend on the particle size. This means that the surface contribution to polarization and conductivity is ignored in MWBH theory. This is also evident from the derivation of eqs (14)–(31), where only bulk properties and partial volumes of components are taken into consideration.

Experiments confirm the MWBH theory of composite dielectrics in a broad range of Φ_2 for many materials (Fig. 2). Nevertheless, there are some factors that are not taken into account in this theory. For example, the spatial distribution of inclusions is assumed to be homogeneous; this is unrealistic in many cases. Aggregation of inclusions affects the properties of a mixture significantly. This effect is considered by Dukhin & Shilov (1974). The second principal problem with MWBH theory is that in real systems the interface between different phases is not as inert as it is assumed to be in eqs (14)–(31).

3.2 Surface effects

It is clear from a thermodynamical viewpoint (Chelidze 1971) that the surface contribution to electrical processes can be quite significant, and even dominating in heterogeneous media

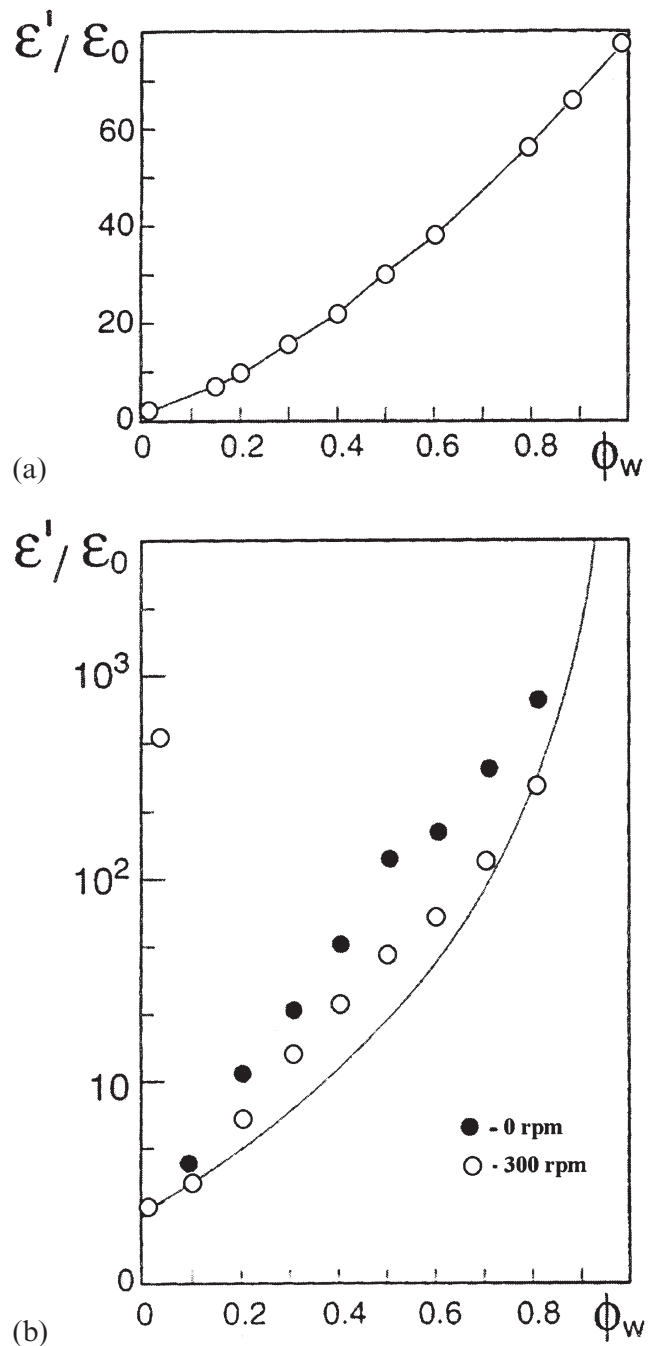


Figure 2. (a) The HF dielectric constant of water in oil emulsions as a function of the dispersed phase (water) content Φ_w at 30 °C (MHz range). The theoretical curve corresponds exactly to the formula of Bruggeman (solid line). Open dots are experimental data (Hanai 1968). (b) The dielectric constant of the same system at 100 Hz at different stirring velocities in rpm: solid dots—at rest; open dots—at 300 rpm. The solid line is the theoretical prediction by the formula of Hanai (1968). The difference in experimental plots for various rpm values comes from the effect of aggregation: at high stirring velocities the emulsion is less aggregated and hence closer to the assumptions of the effective medium theory.

with highly developed interfaces between components. The magnitude of this contribution depends on the following factors: (1) the surface-to-volume ratio of the components; (ii) the surface and volume physical and chemical properties of the

components; (iii) the mode of coupling between surface and volume processes; (iv) the characteristics of the electrical double layer at the solid–liquid interface; and (v) the geometrical, chemical or electrical heterogeneity of real interfaces.

As a rule, many processes develop on surfaces in response to the application of an electric field. At very low water contents almost all water molecules are in an adsorbed state. The properties of adsorbed water differ significantly from those of the bulk liquid: the relaxation time is longer and the energy of activation is higher in the adsorbed state. This means that the properties of adsorbed molecules are intermediate between bulk water and ice (Chelidze *et al.* 1977; Hoekstra & Doyle 1971). The quantitative analysis of dielectric spectra of the first adsorbed layers of water can be performed in the framework of the theory of absolute rates of reactions. The theory allows the estimation of thermodynamic properties of adsorbed water from the experimental data on temperature dependence of relaxation times (Chelidze 1977). The physical mechanism of the polarization of adsorbed layers can be either the orientational polarization of dipole molecules of water or the transfer of charges (protons) along hydroxyl groups or chains, so-called surface charge transfer complexes (Chelidze *et al.* 1977; Pride 1994).

The contribution of surface processes to the ohmic conductivity of porous systems is considered by Dukhin (1975), Pride (1994), Glover *et al.* (1994a,b) and Revil & Glover (1997). In the last three papers a new fundamental theory of ionic surface conductivity is presented which is based on first principles of the physics and chemistry of surfaces.

Surface migration and diffusion also evoke polarization mechanisms, which are often much stronger than the MWBH polarization. The contribution of surface processes to conductivity and polarization can be experimentally estimated by calculating deviations of electrical spectra from MWBH theory.

3.2.1 Geometrical surface effects

Le Mehaute & Crepy (1983), Wong (1987), Ruffet *et al.* (1991a,b), Ruffet (1993) and Glover *et al.* (1994a) considered the geometrical effects resulting from the existence of fractal surfaces. They assumed that the internal surfaces of rocks are rough and examined how the charge layer is built up at the interface. When an electric field is applied along the *Y*-axis normal to the surface, ions have to migrate in this direction. As minerals are mainly dielectrics, ions do not penetrate into them and have to migrate along the rough surface in the *X*-direction in order to progress further in the *Y*-direction.

Assuming that the electrode surface is a self-affine fractal, the time *t* needed to cover the horizontal diffusion distance *r* is $t \sim r^2$. During that time the average displacement *w* along the *Y*-axis is

$$w = r^\beta, \quad (32)$$

where for a self-affine surface $\beta = d - D_f$, D_f being the fractal dimension of the surface and *d* that of the embedding space (i.e. $d = 3$ in our case). At time *t* the polarization is proportional to *w*:

$$P \sim w = r^\beta = r^{d-D_f} = t^{(d-D_f)/2}. \quad (33)$$

The dielectric permittivity has the same frequency dependence as polarization, so introducing the angular frequency ω from

$t \sim \omega^{-1}$ and $d = 3$ the frequency response of ϵ^* is

$$\epsilon^* \sim \omega^{-(3-D_f)/2}. \quad (34)$$

The model predicts a power-law frequency dependence for ϵ with an exponent $(D_f - 3)/2$, and a similar behaviour for σ^* with an exponent $(D_f - 1)/2$. Wong (1987) presented results for the Portland sandstone which agree quite well with these predictions in the megahertz range. In this case the high-frequency behaviour is $\epsilon^* \sim \omega^{-x}$, with $x = 0.29$, so $D_f = 2.42$.

The earlier and more formal model of Le Mehaute & Crepy (1983) gives for a 2-D space an exponent $D_f = 2/(1 - \alpha)$, where α is the Cole–Cole parameter; accordingly, the value of D_f tends to infinity as $\alpha \rightarrow 1$, which makes this approach unrealistic at large α .

For these models, the frequency dependence exponent reflects the distribution of relaxation times, which in turn is connected with ion fluxes passing obstacles of various sizes on the rough surface. This exponent can then be related to the Cole–Cole exponent α . This, in principle, gives a recipe for calculating surface roughness (exponent D_f) from the experimentally determined Cole–Cole parameter α . It should also be noted that according to some numerical studies (Cao *et al.* 1994) surface roughness results in fractal behaviour only at very high frequencies, in the gigahertz range.

Glover *et al.* (1994a) give D_f of saturated sandstones, calculated from experimental HF values of ϵ^* , as a function of fluid salinity and ionic type. These experiments show that purely geometrical fractal surface models of polarization do not fit the experimental data, as it has been found that variations of the electrolyte concentration and cation type change the apparent fractal surface dimension: it decreases with decreasing concentration and increasing ionic radius of the dominant cations, which seems strange because the fractal dimension of the solid surface should be independent of the properties of the electrolyte. The most natural explanation is that the electrical method of assessment of surface roughness is not a direct one; it actually gives apparent ‘electrical roughness’ (Glover *et al.* 1994a). The further from the surface of solid, the less the ‘roughness’ of the equipotential lines of a heterogeneous surface electric field is (Bunde & Havlin 1992). The less the electrolyte concentration is and the larger the ionic radius of the dominant cations in solution is, the further from the grain surface (and hence along a less tortuous route) the transport of dominant charge carriers occurs. This can qualitatively explain the phenomenon of variability of apparent fractal surface dimensions as a result of the multifractal nature (see e.g. Korvin 1992) of electrically measured surface roughness.

3.2.2 Smooth surfaces with homogeneous charges

(a) *Surface polarization: conducting films*: Surface conductivity is due to the high concentration and different mobilities of charge carriers in the electrical double layer (EDL). O’Konsky (1960) calculated the polarization field of spherical particles of radius *a* and bulk conductivity σ_2 . He introduced the frequency-independent surface conductivity σ_s , and assumed that the double-layer ions, in response to the applied electrical field, can move normal to the interface over a distance which is of the order of the Debye length. He concluded that the surface conductivity σ_s causes the same effect on the distribution of the polarization field as an increase in the particle’s

bulk conductivity σ_2 to a value σ_{2e} :

$$\sigma_{2e} = (\sigma_2 + 2\sigma_s/a). \quad (35)$$

Hence, eq. (30) adequately describes the surface conductivity effect if the bulk conductivity σ_2 is replaced by the effective conductivity σ_{2e} . The relaxation time for establishing the polarization is (O'Konsky 1960)

$$\tau = \frac{a(\varepsilon_2 + 2\varepsilon_1)}{8\pi\sigma_s}, \quad (36)$$

where ε_1 and ε_2 are the DP of the electrolyte and the particle respectively.

Schwarz (1962) allowed for the frequency-dependent surface conductivity of bound ions within the double layer. In his model, the application of an electric field resulted in only tangential migration of charges; displacement normal to the interface was forbidden. In this case, the equilibrium is reached through diffusion of ions along the interface. The relaxation time for attaining equilibrium is then obtained as a typical diffusion time constant (Wert & Thomson 1964; Dukhin & Shilov 1974):

$$\tau_s = \frac{a^2}{2D_s}, \quad (37)$$

where D_s is the surface diffusion coefficient of bound ions.

A general solution to the problem of polarization of a non-conducting particle of radius a and dielectric permittivity ε_2 coated with a film of surface conductivity σ_s and surface diffusion coefficient D_s and embedded in a medium with complex dielectric permittivity ε^* in an ac field leads to the following expression for the oscillating dipole moment d_e of the particle (Dukhin *et al.* 1969; Chelidze *et al.* 1977):

$$d_e = \frac{a^3}{\varepsilon_2 + 2\varepsilon_1} \left[\varepsilon_2 + \varepsilon_1 + 3\varepsilon_1 \frac{1 - (\tau/\tau_s)}{1 + i\omega\tau} \right] E, \quad (38)$$

where

$$\tau = \frac{\tau_s}{1 + \frac{4\pi a\sigma_s}{D_s(2\varepsilon_1 + \varepsilon_2)}}. \quad (39)$$

For small particles the second term in the denominator of eq. (39) is small, and the relaxation is controlled by diffusion along the surface $\tau = \tau_s$. For large particles the second term dominates and we obtain O'Konsky's relaxation time, eq. (36).

At low frequencies ($\omega\tau \ll 1$) and for high σ_s , a dielectric particle coated with a conducting film and embedded in a non-conducting medium is polarized as a conducting inclusion. This results in $\varepsilon \rightarrow \infty$ when $\Phi_2 \rightarrow 1$, according to eq. (31).

This explains why the addition of small quantities of water to a porous rock causes a strong dielectric increment at low frequencies, much larger than that expected from MWBH theory or from the orientational polarization of adsorbed molecules (Chelidze *et al.* 1977).

(b) *Surface polarization: insulating films*: The case of insulating films covering conducting particles is the reverse of the previous case. It corresponds to a system such as oxidized metal inclusions dispersed in a conductive medium. Such a situation could be observed with metallic ores.

The corresponding theoretical model has been developed by Pauly & Schwan (1959). They modified the two-component

effective medium theory by taking into account the existence of a third component, an insulating envelope of thickness h with dielectric permittivity ε_3 and conductivity σ_3 coating spherical particle of radius a with electrical parameters ε_2 and σ_2 . The envelope is assumed to be thin, $h \ll a$. The coated particles are embedded in a medium with characteristic parameters ε_1 and σ_1 . The equation for a complex DP is

$$\varepsilon^* = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_i}{1 + i\omega\tau_p} + \frac{\varepsilon_i - \varepsilon_\infty}{1 + i\omega\tau_q} - \frac{i\sigma_0}{\omega}, \quad (40)$$

where $\varepsilon_s, \varepsilon_i, \varepsilon_\infty$ are functions of components ε and σ , σ_0 is the dc conductivity of the system, τ_q is the relaxation time for a suspension of non-coated particles and τ_p is that for a suspension of coated particles, $\tau_q < \tau_p$ (Pauly & Schwan 1959). Since there are two relaxation times there are accordingly two 'static' values of dielectric permittivity—one at very low frequencies (ε_s) and the second in the intermediate-frequency range (ε_i).

Generally, $\sigma_3 \ll \sigma_1$, $\sigma_3 \ll \sigma_2$ and $h \ll a$. After the introduction of coating conductance $G_s = \sigma_3/4\pi h$ and its capacitance $C_s = \varepsilon_3/4\pi h$, eq. (40) simplifies greatly in the LF limit, where in most cases the condition $G_s \ll \sigma_2/(a+h)$ is fulfilled and we obtain

$$\sigma = \sigma_1 \frac{2(1 - \Phi)}{2 + \Phi}, \quad (41)$$

$$\varepsilon_s - \varepsilon_i = 4\pi \frac{9\Phi a C_s}{(2 + \Phi)^2},$$

$$\tau_p = (a+h)C_s \left(\frac{1}{\sigma_2} + \frac{1}{\sigma_1} \frac{1 - \Phi}{2 + \Phi} \right),$$

where Φ is the partial volume of coated particles and $\Phi \cong \Phi_2$, because the envelopes are negligibly thin.

Such systems undergo strong frequency dispersion of ε and σ , which is explained by the shunting of insulating films at sufficiently high frequencies. Shunting incorporates the bulk conducting volume of particles into the global conduction process, causing an increment of conductivity and a drastic decrease of the dielectric constant to MWBH values at high frequencies.

(c) *Open electrical double layer*: The 'gigantic low frequency polarization' theory (GLFP) considers transient processes in the bulk of the pore electrolyte caused by polarization of the electrical double layer (EDL) at the solid-liquid interface. The potential ψ within the EDL is shown in Fig. 3, where the total potential drop ψ_0 between the solid and the bulk of the electrolyte includes that in the diffuse, mobile part of the EDL, which is usually called the ζ potential, and that in the dense, immobile part of the EDL, which in turn includes the Stern potential ψ_s and the potential drop $(\psi_0 - \psi_s)$, due to chemically bound non-solvated ions which cannot be exchanged with the electrolyte. It is expected that all ions which are at a potential field $\psi < \psi_s$ can be exchanged with the bulk of the electrolyte.

Thus, unlike the Schwarz approach, the GLFP theory allows for a free exchange of EDL excess (transient) charges with the electrolyte, which is accomplished through the displacement of ions normal to the interface. This makes the polarization process much more effective. Application of an electric field to a charged particle embedded in an electrolyte leads first to a distortion of the initially symmetrical EDL due to migration

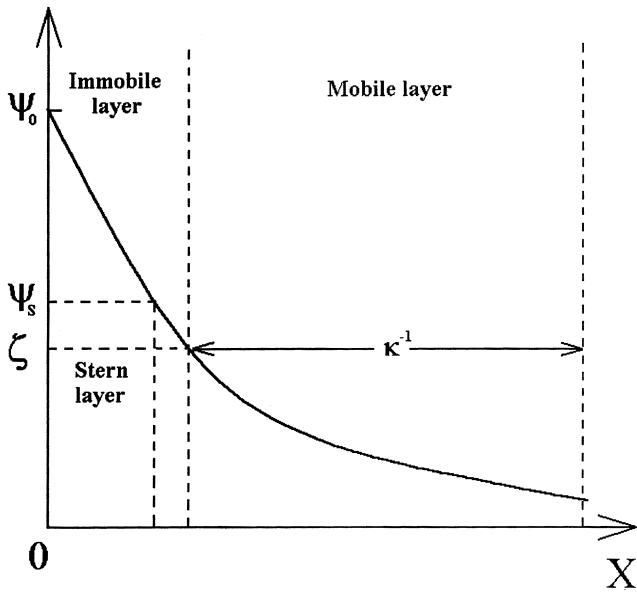


Figure 3. Surface potential ψ versus distance from the solid surface x . ψ_0 is the total potential drop between the surface of the solid and the bulk of electrolyte, ψ_s and ζ are respectively potential drops in the Stern layer and in the mobile part of the EDL (modified from Adamson 1982).

and diffusion of charges in it. In the following treatment only sufficiently large particles will be considered, satisfying the condition $a \gg \kappa^{-1}$, where a is the particle radius and κ^{-1} is the Debye screening length or thickness of the diffuse double layer. As the charge carriers in EDL are transported in a different ratio to those in the bulk electrolyte, and as the surface fluxes of anions and cations are not limited to flow parallel to the interface as in previous models, the asymmetry of anion and cation fluxes and the electroneutrality condition for a pore electrolyte result in the appearance of concentration gradients near the particle, outside the EDL, in addition to the electric polarization field of the deformed EDL. We can analyse this behaviour through a sequence of 4 steps (Fig. 4). The relaxation time for establishing a steady-state polarization of EDL of a negatively charged particle (Fig. 4a) is small: several microseconds for brine conductivities of about 10^{-4} S m^{-1} , and this corresponds to the first step. Due to the free exchange between the open EDL and the bulk electrolyte, the non-equilibrium concentration of charges, developed in EDL, results in the appearance of local transient clouds with the excess of positive charges on one side of the particle and with their deficit on the other side within the bulk electrolyte: this is the second step (Fig. 4b). The third step (Fig. 4c) involves the time τ_{dl} needed to attain the electrical equilibrium state between charged diffusion clouds and the adjoining volume of macroscopically neutral bulk electrolyte. This time is also small because the charges diffuse only over a distance κ^{-1} , so (Dukhin & Shilov 1974)

$$\tau_{dl} = \frac{1}{\kappa^2 D_{ef}}, \quad (42)$$

where

$$D_{ef} = \frac{D^+ D^- (z^+ + z^-)}{z^+ D^+ + z^- D^-}, \quad (43)$$

z^- and z^+ are valences and D^- and D^+ are diffusion coefficients of anions and cations. Hence at frequencies $\omega \ll 2\pi/\tau_{dl}$, the bulk electrolyte near the EDL regains its electroneutrality, and only asymmetry in concentration remains from charged clouds outside the EDL. However, steady-state polarization due to the induced concentration gradients (the last step, Fig. 4d) is established over a much longer interval of time, as it is controlled by diffusion over distances of the order of the linear dimension of the particle. The relaxation time τ_a in this case is the time needed for a diffusion front to propagate over a distance a (Dukhin & Shilov 1974):

$$\tau_a = \frac{a^2}{2D}, \quad (44)$$

where D is the bulk diffusion coefficient of the electrolyte. Because of this, the polarization field of the particles lags behind the applied electric field at low frequencies.

The key point is that in a conducting medium the polarization field induces conduction currents which lag behind the applied field. These out-of-phase currents appear as displacement currents. Such large displacement currents are seen experimentally as high values of the dielectric constant. Thus, on the one hand a large electric charge of a particle produces a significant asymmetry of ion fluxes and diffusion gradients, and on the other hand a large electrolyte conductivity produces large out-of-phase currents. Only together can these effects result in a large low-frequency dielectric constant, which can be considered as an effective parameter of a system. It can be shown (Dukhin & Shilov 1974; Sen & Chew 1984) that high values of DP are obtained only when the surface charge is sufficiently large and $\kappa a \gg 1$. If $\kappa a \ll 1$, the dielectric constant of the mixture cannot exceed the MWBH values of ϵ (Trukhan 1962).

A simplified version of the GLFP theory can be written as follows. From Maxwell–Wagner theory, we use eq. (27), which we rewrite in terms of the dipole moment of a spherical particle d_e^* and the number of particles per unit volume N . The partial volume of inclusions, Φ_2 , for spherical particles of radius a is $4\pi a^3 N/3$, thus eq. (27) can be written in complex form as

$$\Delta \epsilon^* = 4\pi \epsilon_1^* N d_e^* / E, \quad (45)$$

where d_e^* is the complex dipole moment of a polarized sphere,

$$d_e^* = \frac{\epsilon_2^* - \epsilon_1^*}{2\epsilon_1^* + \epsilon_2^*} a^3 E. \quad (46)$$

The real part of the complex dielectric permittivity from eq. (45) is

$$\text{Re } \Delta \epsilon^* = 4\pi N [\text{Re}(\epsilon_1^*) \text{Re}(d_e^*/E) + \text{Im}(\epsilon_1^*) \text{Im}(d_e^*/E)]. \quad (47)$$

Here, $\text{Im}(\epsilon_1^*)$ is the imaginary part of the electrolyte complex dielectric permittivity. It increases greatly when frequency decreases, and varies as σ/ω . Hence, if $\text{Im}(d_e^*/E)$ is sufficiently large, the product $\text{Im}(\epsilon_1^*) \text{Im}(d_e^*/E)$ can be very large at low frequencies, so the value $\text{Re } \Delta \epsilon^*$ as a whole becomes large.

It seems that the above model provides a universal explanation of the amplification of polarization of any origin in a moderately conducting media.

A comprehensive analysis of the processes in EDL involves the solution of the equations of convective diffusion, Poisson and Navier–Stokes equations with the assumption that κa is large. This corresponds to the case of a moderate electric field which means that the Boltzmann's distribution holds in EDL.

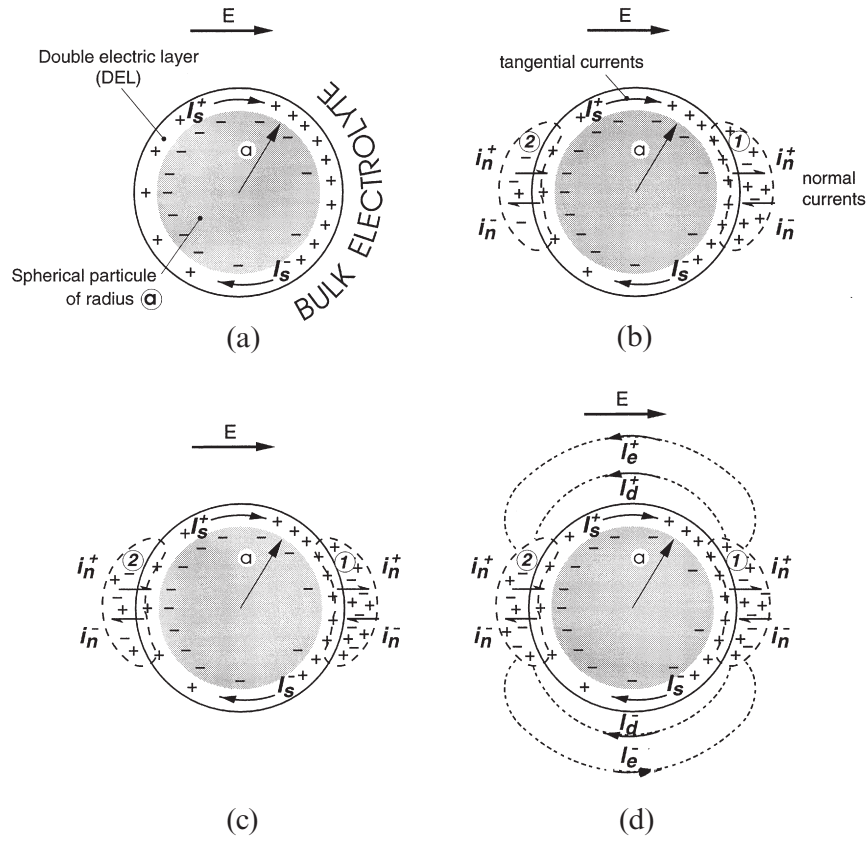


Figure 4. Model of evolution of EDL polarization for a negatively charged spherical particle of radius a under an electric field E . The particle is immersed in the electrolyte and EDL is present on its surface. (a) Tangential fluxes of positive (I_s^+) and negative (I_s^-) ions in the double layer result in an asymmetric charge distribution within the EDL. (b) Normal fluxes of anions and cations (i_n^+ , i_n^-) between the EDL and the bulk electrolyte result in high- and low-concentration areas or diffuse charged clouds on both sides of a particle in the bulk electrolyte. Electroneutrality is not achieved at this stage. (c) Electroneutrality is achieved within a short time interval because the time constant $\tau_{dl} = 1/\kappa^2 D_e$ in both diffuse clouds. (d) Concentration differences and electrical potential differences between areas (1) and (2) generates both diffusion (i_d^+ , i_d^-) and migration (i_e^+ , i_e^-) fluxes of anions and cations which lag behind the applied field E . The time constant of this process is large, $\tau = a^2/D$.

The dielectric increment in the static limit $\Delta\epsilon_s$ for highly (negatively) charged particles in an electrolyte with $z^+ = z^- = z$ is (Dukhin & Shilov 1974; Chelidze *et al.* 1977)

$$\Delta\epsilon_s = \frac{9\Phi_2\epsilon_1 \left[3m \exp\left(\frac{z|\zeta'|}{2}\right) + \exp\left(\frac{z|\psi'_s|}{2}\right) \right]^2}{4z \left[1 + 3m \left(\frac{1}{z} + \frac{1}{z^2} \right) \frac{\exp\left(\frac{z|\zeta'|}{2}\right)}{\kappa a} + (z+1) \frac{\exp\left(\frac{z|\psi'_s|}{2}\right)}{\kappa a} \right]^2}, \quad (48)$$

where Φ_2 is the partial volume of inclusions, $\kappa a \gg 1$:

$$m = \frac{\epsilon_1 \left(\frac{kT}{e} \right)^2}{6\pi\eta D^+}, \quad (49)$$

$$\zeta' = ze\zeta/kT, \quad (50)$$

$$\psi'_s = ze\psi_s/kT. \quad (51)$$

Here, k is the Boltzmann constant, a is the particle size, T is the absolute temperature, D^+ is the diffusion coefficient of cations, η is the dynamic viscosity of the fluid, ζ is the potential

of the shear plane, ψ_s is the Stern potential, e is the electron charge and z is the valence of anions. The value of m is about 0.1 for aqueous suspensions.

The maximum value of $\Delta\epsilon_s$, attained at $\kappa a \gg 1$, is

$$\Delta\epsilon_s^{\max} = \frac{9\Phi_2\epsilon_1}{4z} \left(3m \exp\left(\frac{z|\zeta'|}{2}\right) + \exp\left(\frac{z|\psi'_s|}{2}\right) \right)^2. \quad (52)$$

Thus, if the particle is large enough, the static DP does not depend on their size.

In order to check this theory quantitatively, complex dielectric constant measurements and surface charge determinations by electrokinetic methods were carried out simultaneously on a suspension of negatively charged spherical non-conducting particles with relatively smooth homogeneous surfaces (latex spheres). The frequency range is 300 Hz–100 kHz. The experimental results are presented in Table 1.

It can be seen from Table 1 that the ζ potential values, corrected for the retardation of the ionic atmosphere of particles (ζ'_{cor}), obtained from electrophoretal measurements are close the values of Stern potential ψ'_s , calculated by GLFP theory from (48). This means that ion fluxes from the immobile part of EDL are insignificant. This agrees with the results obtained by Revil & Glover (1997). The average particle radii, calculated from the critical frequency using (44), are also in

Table 1. Comparison of dielectric, electrophoretic and microscopic measurements on the model system: suspension of spherical particles in an electrolyte (Chelidze *et al.* 1977).

Substance	Φ_2	τ_a , s	a_d , μm	a_m , μm	κa_d	ζ'_m	ζ'_{cor}	ψ'_s	$\Delta\epsilon_m$	$\epsilon_\infty/\epsilon_0$
1	0.02	8×10^{-6}	0.13		15	2.30	3.50	3.50	68	76
1	0.04	8×10^{-6}	0.13		15				136	75
1	0.08	8×10^{-6}	0.13		15				272	70
1	0.18	8×10^{-6}	0.13		15				612	57
2	0.05	10^{-3}	0.45		60	2.60	3.00	3.20	700	76
3	0.10	5×10^{-6}	0.10		20	2.90	4.00	3.80	780	66
3	0.025	5×10^{-6}	0.10	0.09	20				120	75
3	0.013	6×10^{-4}	1.10	1.60	95	3.00			9750	76

Substances are: 1—chlorophren latex, 2—aged chlorophren latex, 3—polyvinylchlorid latex; Φ_2 is the volume fraction of inclusions; τ_a is the relaxation time from dielectric spectra in s; a_m is the particle radius value determined from microscopic measurements in μm ; a_d is the particle radius predicted by eq. (44) in μm ; κa_d is the product of the inverse Debye screening length and particle radius; ζ'_m is the reduced value of the measured ζ potential (see eq. 50); ζ'_{cor} is a value of ζ'_m corrected for the additional retardation of particles at electrophoresis resulting from the relaxation of their ionic atmosphere (Dukhin & Shilov 1974); ψ'_s is the reduced Stern potential or total potential drop in both mobile and immobile parts of EDL; $\Delta\epsilon_m$ is the experimental (relative) dielectric increment, i.e. the difference between LF and HF values of the relative dielectric constant; ϵ_∞ is the relative dielectric constant attained at high frequencies. The ohmic conductivity of all samples was of the order of 10^{-2} S m^{-1} .

agreement with independent electron microscopy measurements. Thus both relaxation times and the magnitude of the polarization $\Delta\epsilon_m/\Phi_2$ conform GLFP theory for this simple model system.

3.2.3 Surfaces with non-homogeneous charges

In the Dukhin–Shilov theory it is implicitly assumed that the surfaces of particles are homogeneously charged. This condition is evidently fulfilled in latex suspensions where surface charges are distributed uniformly. The surfaces of minerals, however, are electrically heterogeneous. This can result from surface roughness, for example, because the asperities can be considered as small charged particles glued to the surface. Moreover, many studies have shown that even very finely polished surfaces of minerals are electrically heterogeneous. Mineral surfaces are actually mosaics of negatively and positively charged areas. The heterogeneity of the surface depends strongly on the mineral type and the pH of the solution (Revil & Glover 1997).

Chelidze *et al.* (1977) proposed a model of polarization of electrically heterogeneous surfaces of solids in which clusters of charges are considered as analogues of charged particles in the GLFP theory. The relaxation time in this model is controlled by the mean size of charged surface clusters a_{cl} and not by particle size a :

$$\tau_{\text{cl}} = a_{\text{cl}}^2 / 2D, \quad (53)$$

and the LF dielectric increment can be approximately calculated from eq. (48) using a_{cl} instead of a and Φ_{cl} instead of Φ_2 . Here Φ_{cl} is the concentration of surface clusters in the unit volume $\Phi_{\text{cl}} \sim (S_{\text{cl}}/S)\Phi_2$, where S is the surface area of the particle and S_{cl} is the mean surface area of the cluster.

A similar model was advanced earlier by Marshall & Madden (1959). These authors described a two-mineral mixture where mineral surfaces are a mosaic of areas which are either covered by clay (zone 1) or free of it (zone 2). In zone 1 the EDL contains a large number of cations (or anions); zone 2 is depleted in them. Application of an electric field generates a concentration gradient around zones. The relaxation time τ

needed to attain equilibrium is $\tau \propto L_2$, where L is an average distance between clay islands.

The clay coating model (De Lima & Sharma 1992; Garrouch & Sharma 1994) is also similar to GLFP theory; here the clay envelope plays the role of a polarized EDL. The relaxation time derived in this case is identical to expression (44).

3.3 Partial saturation effects: percolation models

The MWBH theory assumes that the embedding media can be filled by inclusions up to high values of Φ_2 homogeneously, without clustering of particles. However, experiments show that the random addition of conductive inclusions to an insulating matrix leads to the formation of clusters of conductors, and eventually to a transition from insulating to conducting behaviour at some critical concentration of the conducting phase Φ_c . This critical threshold is reached when a continuous conducting path is formed in the system. Gueguen *et al.* (1997) showed that the critical concentration, predicted by effective medium theory, results from an attempt to apply the theory beyond its limits of validity, and does not correspond to real data.

Conductor–insulator transitions are correctly described by percolation theory (Broadbent & Hammersley 1957; Shante & Kirkpatrick 1971; Stauffer & Aharony 1992; Sahimi 1994; Gouyet 1992). The critical concentration is defined as a percolation threshold and the system-spanning cluster of conductive inclusions as an infinite cluster (IC). This approach is obviously of great interest for rocks, which are mixtures of conductors and insulators.

Shante & Kirkpatrick (1971) developed a self-consistent model in order to calculate the conductivity of a network of bonds with random conductances. Percolation theory predicts the behaviour of ϵ and σ as a power function of the quantity $\Delta\Phi = (\Phi - \Phi_c)$ in the critical region of conductor concentration Φ , namely, near the percolation threshold.

Geometrically, bond percolation can be defined on a regular lattice of bonds, which are either occupied with probability p , or empty with probability $(1 - p)$. A set of connected occupied bonds forms a cluster. Clusters are of finite size at $p < p_c$ and

form an infinite cluster at $p = p_c$. As percolation is similar to critical transitions the 'distance' Δp from the critical point p_c is the most important parameter in percolation theory:

$$\Delta p = p - p_c. \quad (54)$$

The percolation probability $P(p)$ defines the probability that a given bond belongs to the IC. $P(p) = 0$ at $p < p_c$, but at $p > p_c$, percolation probability P increases as

$$P(p) \sim \Delta p^\beta, \quad (55)$$

where β is the critical exponent of percolation probability.

The second important parameter is the correlation length L_c , which can be loosely defined as the maximal size of finite clusters of occupied bonds for $p < p_c$. A similar definition is true for the correlation length of empty bonds above the percolation threshold ($p > p_c$). The correlation length diverges as $p \rightarrow p_c$:

$$L_c(p) \sim \Delta p^{-\nu}, \quad (56)$$

where ν is the critical exponent of correlation length.

The occupied bonds can be identified as conductors of conductance b and the empty ones as insulators of conductance a . Thus geometrical percolation maps on to the problem of the electrical conductivity of composites. The electrical properties of such systems are completely defined by the ratio $b/a = h_c$ and the increment of p , i.e. by the parameter Δp . The critical behaviour of conductivity near the percolation threshold allows the introduction of the following scaling expressions for the effective conductivity of the network, σ (Efros & Shklovskii, 1976; Sahimi 1994):

$$\sigma = a|\Delta p|^t f_\pm(x), \quad (57)$$

where

$$x = h_c|\Delta p|^{-t-s} \quad (58)$$

and t and s are the critical exponents of conductivity for the transitions metal–insulator and conductor–superconductor respectively, and f_+ and f_- are homogeneous functions, corresponding to $p > p_c$ and $p < p_c$ respectively.

The most interesting contribution of the percolation approach to the theory of composites may be the prediction of scale dependence of σ near the percolation threshold. In contrast to effective medium theories, which imply the existence in a heterogeneous system of a representative elementary volume (REV), percolation theory predicts that at $p = p_c$ the effective conductivity depends on the sample size in a regular manner (Havlin & Bunde 1991):

$$G \sim L^{-t}, \quad (59)$$

where L is the size of the sample and t is the conductivity exponent. The point is that near the percolation transition the system is fractal and contains non-conducting clusters (holes). The size of the holes increases as the system approaches the threshold from above p_c , causing a regular decrease of σ . On the other hand, in finite conducting systems at $p > p_c$ the fragmentation of the sample will cause a transition from a conducting to an insulating state when the scale of partition becomes less than the correlation length of non-conducting clusters L_{cn} ; in most cases, for a small part of the system the conductivity path present in the large sample will be destroyed. This means that the hypothesis of REV, which is the basic assumption of the effective medium theories, is not valid for

systems near percolation transitions. Scale dependence of specific physical properties, obtained from the first principles of modern theory of disordered media, is a striking result, which may even affect the interpretation of geophysical data (Jones 1995; Chelidze 1987).

Eq. (57) can be generalized for complex conductances a^* and b^* using either lattice gas or equivalent circuit models (Efros & Shklovskii 1976; Havlin & Bunde 1991; Sahimi 1994). Here we follow the latter approach. If p is the fraction of bonds which are purely resistive, where R is the resistance of the bond and $(1 - p)$ is the fraction of purely capacitive bonds, where the capacitance of the bond is C , then, according to Sahimi (1994),

$$a = 1/R, \quad (60)$$

$$b = i\omega C. \quad (61)$$

The conductance ratio $h_c = b/a$ is given by

$$h_c = i\omega/\omega_0, \quad (62)$$

where $\omega_0 = 1/RC$.

In the static limit $\omega = 0$, the capacitors become insulators and the model reduces to a conductor–insulator mixture; in this case the σ^* is calculated by eq. (57), where

$$a = 1/R, \quad (63)$$

$$x = i(\omega/\omega_0)|\Delta p|^{-t-s}. \quad (64)$$

The frequency-dependent complex dielectric permittivity is obtained as follows:

$$\varepsilon^* = \sigma^*/i\omega. \quad (65)$$

This equation predicts unusual behaviour of the static DP as $\omega \rightarrow 0$. Using a Taylor expansion of $f(x)$ around $x = 0$ (Clerc *et al.* 1990; Sahimi 1994), we obtain from (65) and the complex analogue of eq. (57)

$$\varepsilon \sim A|\Delta p|^{-s}, \quad (66)$$

where A is the amplitude factor. This equation predicts that ε diverges when $p \rightarrow p_c$. This effect is called polarization catastrophe. Eq. (66) is similar to eq. (31), but there is a difference in the critical concentration value and the exponent, which is explained by a different approach to the structure of the composite: the former uses a fractal approximation and the latter a continuous media approximation.

The third striking result is for loss angle. As $p \rightarrow p_c$, the effective conductivity becomes

$$\sigma^* \approx \frac{f(0)}{R} \left(\frac{i\omega}{\omega_0} \right)^{t+t+s}. \quad (67)$$

This means that the frequency dependences of both σ' and σ'' are identical. Then, as $p \rightarrow p_c$, the ratio $\sigma'/\sigma'' = \varepsilon''/\varepsilon' = \tan \delta$ is independent of frequency (Fig. 5a); that is, the loss angle in the critical state is universal ($\delta_c = \text{const}$):

$$\delta_c = \frac{\pi}{2} \frac{s}{t+s}, \quad (68)$$

and for 3-D systems it should be equal to 0.45. The Cole–Cole plot is asymmetric and at $p = p_c$ degrades to a straight line with a slope of 45°. It may be difficult to reveal these peculiarities experimentally because, as p deviates from p_c , the unusual behaviour of δ is observable only in a limited frequency range.

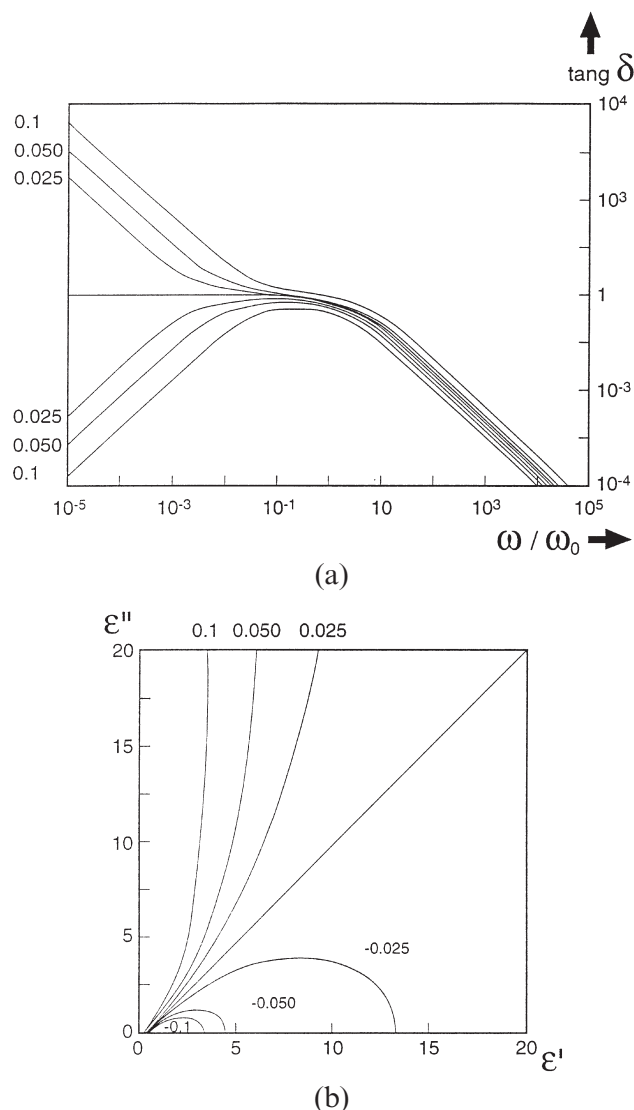


Figure 5. (a) A log-log plot of $\tan \delta$ versus reduced frequency ω/ω_0 according to the effective network approximation (ENA) for the 3-D RC model on a cubic lattice. Values of distance from the percolation threshold Δp are indicated on the curves. (b) Cole-Cole plot of the complex dielectric constant according to ENA for the same model; values of Δp are indicated (Clerc *et al.* 1990).

The percolation model can be useful in interpreting interesting hysteresis phenomena in electrical and other transport properties of partially saturated porous rocks (Knight & Nur 1987; Knight 1991) because it takes into consideration not only the content of water but also the details of fluid distribution in the porous space, which can be different at imbibition and drainage for the same water content. It seems that the site-bond percolation model is an appropriate tool for the explanation of the hysteresis effect (Paper II).

We show in Paper II how the above theories may be applied to real rocks with various water contents.

4 CONCLUSIONS

Electrical spectra of porous rocks in the 1 mHz–10 MHz range reflect numerous polarization processes resulting mainly

from rock heterogeneity. The first is the Maxwell–Wagner polarization due to differences in the electric bulk properties of rock components. Maxwell–Wagner theory allows the prediction of electrical spectra of mixtures from bulk partial volumes, the properties of their components and their microstructure. However, Maxwell–Wagner theory, even when extended further by Bruggeman for concentrated systems and by Hanai for the complex dielectric constant or complex conductivity of components, proves unable to explain the anomalously large polarization in water-bearing rocks at low frequencies. The problem is that the MWBH theory does not take into account important points, such as surface conductivity and polarization, as well as clustering of components in disordered mixtures. Nevertheless, the MWBH theory is a starting point for any theoretical interpretation of experimental electrical spectra.

The simplest surface effect is due to surface migration and diffusion of charges under an applied field limited by a thin surface layer; this can be a conductive film or a closed electrical double layer. Film polarization is especially strong at moderate partial saturations, where it can produce strong LF polarization.

Another three-component system, a conducting matrix containing conductive grains with a thin insulating envelope, also shows a strong LF polarization. An example of such a composite is a system of oxidized metal particles embedded in brine-saturated rock. In this case, polarization is due to the large capacitance of thin dielectric shells, which dies out at high frequency when shells are shunted by displacement currents.

Very strong dielectric effects are caused by the polarization of ‘open’ electric double layers on the solid–liquid interface at large saturations. Whereas polarization of bound charges in ‘closed’ double layers produces relatively weak polarization, akin to that of films, diffusion potentials generated by the free exchange of ions between ‘open’ deformed EDL and the adjoining electrolyte produce out-of-phase currents, and hence large values of effective dielectric constant. This model is known as GLFP; it can also be applied to heterogeneously charged surfaces.

The anomalous dielectric properties of partially saturated rocks can alternatively be interpreted using percolation theory. This theory predicts that, when the conductive fraction increases, clustering of conductive inclusions develops, and the thickness of insulating gaps between conductive clusters decreases, causing a large increment in the capacitance of the sample. The dielectric catastrophe is expected near the percolation threshold where the dielectric constant diverges. Further increases in the conductive component cause the shunting of insulating capacitive gaps and the decrease of the dielectric constant. The difference amongst the equations which describe the effects of dielectric catastrophe in MWBH and percolation theories illustrates the difference between continuous and fractal approaches to disordered media. The fractal approach also allows an understanding of the scale dependence of specific electric properties of heterogeneous materials; this is not possible in MWBH theory.

The multitude of possible relaxation processes and available theories call for model experiments on simple systems under controlled conditions using independent methods in addition to electrical spectroscopy. Experimental electrical spectra of real geological objects and geophysical applications are considered in the Paper II.

As the electrical spectra are actually Fourier transforms of the time variation of polarization decay, all the theories quoted can be applied to the interpretation of both ac and induced-polarization data.

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APPENDIX A: SYMBOLS AND ABBREVIATIONS

- a —radius of particle
 a_{cl} —characteristic size of charged surface cluster
 C —capacitance
 d —Euclidian dimension
 d_e —dipole moment of particle
 D —bulk diffusion coefficient of electrolyte

- D_e —electrical induction
 D_f —fractal dimension
 D_s —surface diffusion coefficient
 E —electrical field intensity
 f —frequency (Hz)
 f_c —critical frequency
 f_1, f_2 —field ratio in composite dielectric
 h —conductivity ratio in composite dielectric
 J —electrical flux
 k —Boltzmann constant
 L —sample size
 L_c —correlation length in percolation theory
 p —probability of site (bond) occupation in percolation theory
 p_c —probability p at percolation threshold
 P —polarization
 P —percolation probability
 R —resistance
 s —critical index of conductor–superconductor transition in percolation theory
 t —conductivity critical index in percolation theory
 T —temperature, °C
 z —valency
 Z —impedance
- α —Cole–Cole exponent
 β —critical exponent of percolation probability
 δ —loss angle
 ϵ —dielectric permittivity
 ϵ_0 —dielectric permittivity of vacuum
 ϵ_d^* —complex dielectric permittivity excluding ohmic conductivity
 ϵ_r —relative dielectric permittivity (dielectric constant); $\epsilon_r = \epsilon/\epsilon_0$
 ϵ^* —complex dielectric permittivity
 ϵ_s —static dielectric permittivity
 ϵ_∞ —high-frequency (unrelaxed) dielectric permittivity
 ϵ' —real part of complex dielectric permittivity
 ϵ'' —imaginary part of complex dielectric permittivity, including σ
 ϵ_d'' —imaginary part of complex dielectric permittivity, excluding σ
 ϵ_1 —dielectric permittivity of embedding substance in composite
 ϵ_2 —dielectric permittivity of inclusions
 ϵ_3 —dielectric permittivity of shell
 η —dynamic viscosity
 κ^{-1} —Debye screening length
 ν —critical index of correlation length
 ρ —specific electrical resistivity
 σ —specific conductivity
 σ_0 —dc conductivity
 σ^* —complex conductivity
 σ' —real part of complex conductivity
 σ'' —imaginary part of complex conductivity
 σ_1 —conductivity of matrix
 σ_2 —conductivity of inclusions
 σ_3 —conductivity of shell ($S\ m^{-1}$)
 σ_s —specific surface conductivity (S)
 σ_e —brine conductivity
 τ —relaxation time
 τ_a —relaxation time of particle of radius a
 τ_{cl} —relaxation time of charged surface cluster

τ_s —relaxation time for surface diffusion of particle of radius a
 Φ_2 —partial volume of inclusions
 Φ_c —critical partial volume of inclusions
 ω —angular frequency
 ω_c —critical angular frequency
 ψ —surface potential at distance x from surface
 ψ_s —Stern potential

$\mathcal{I}m$ —imaginary part
 $\mathcal{R}e$ —real part
 EDL—electrical double layer
 DP—dielectric permittivity
 DC—dielectric constant
 MWBH—Maxwell–Wagner–Bruggeman–Hanai
 GLFP—gigantic low-frequency polarization